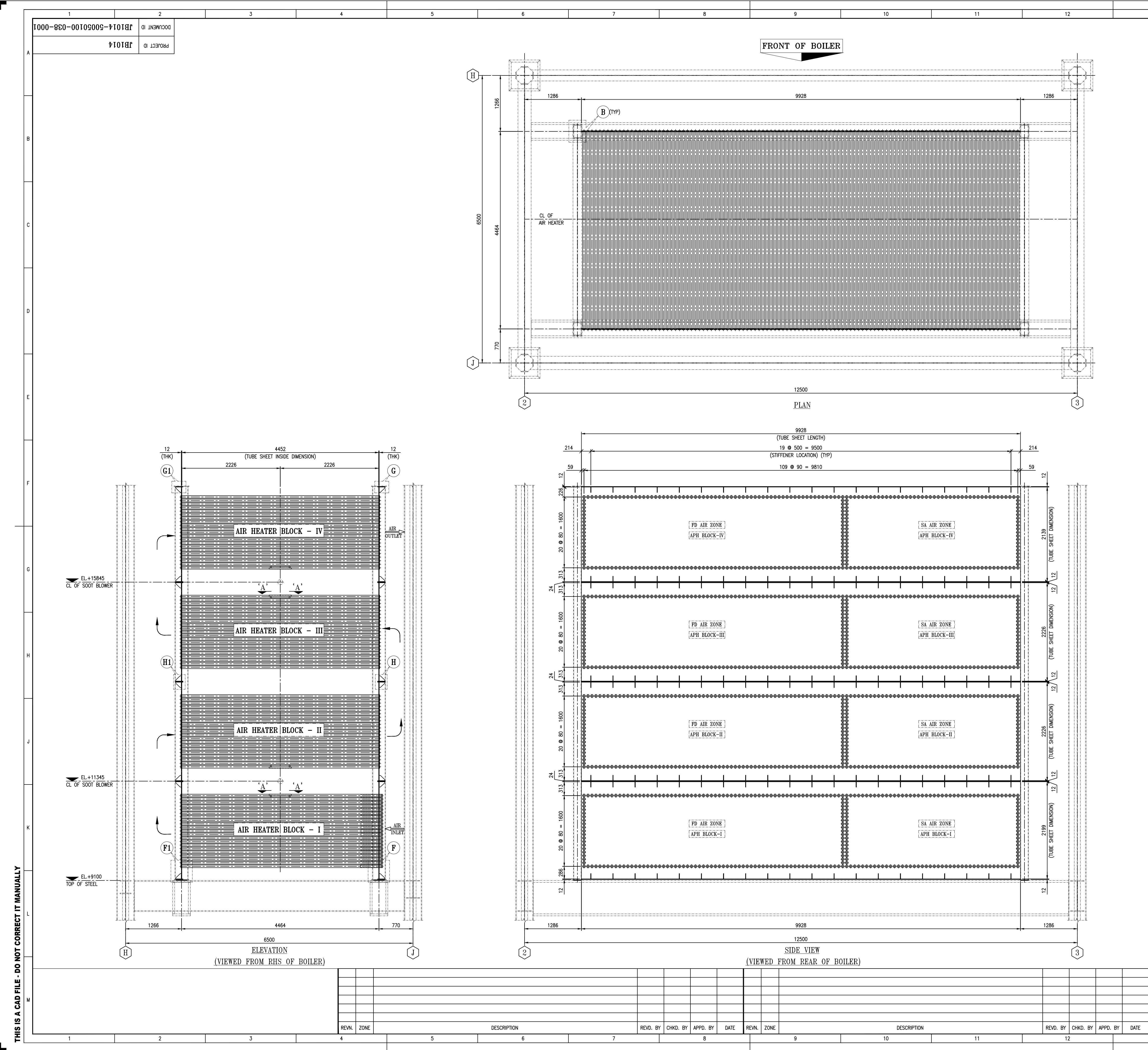




NOTES:-

01. ALL DIMENSIONS ARE IN mm UNLESS OTHERWISE SPECIFIED.
02. HOLE TOLERANCE OF ± 0.4 TO BE STRICTLY MAINTAINED.
03. FABRICATION TO BE DONE BY FULL SCALE SHOP LAYOUT.
04. TUBE HOLE & TUBE ENDS ARE TO BE THOROUGHLY CLEANED BEFORE EXPANSION AT SITE.
05. I/S - DENOTES INSIDE DIMENSIONS, FW - DENOTES FIELD WELD.
06. WELDING ELECTRODE SHALL CONFORM TO IS:814 (LATEST).
07. ALL WELDING SHALL BE DONE AS PER IS:816 & 823 & SHALL BE FULL CONTACT LENGTH UNLESS STATED OTHERWISE.
08. ALL STRUCTURAL MEMBERS CONFORM TO IS:808.
09. SURFACE FINISH OF TUBES (50mm AT BOTH ENDS) TO BE MAINTAINED AS 120 MICRONS INCHES.
10. FOR DIMENSIONAL TOLERANCE REQUIREMENT PLEASE REFER DOC.No. GEN-50990100-DSM-002 AND ITS ASSOCIATED DRAWING.
11. AIR HEATER TUBES (ITEM CODE : AHT001, AHT002), SLEEVE (ITEM CODE : AHT003) ARE RELEASED AS BOUGHT OUT ITEMS ALONG WITH OUR AIR HEATER TUBE SPECIFICATION JB1014-50050100-SPC-0001.
12. BOTH TUBE SHEET OF EACH AIR HEATER BLOCK TO BE CLAMPED TOGETHER AND STACK DRILLED TO ENSURE THAT ALL TUBE HOLES COINCIDE.

PAINTING NOTES:-

FOR PAINTING NOTES REFER SPECIFICATION No: JB1014-50570900-SPC-0001.

ERECTION NOTES:-

01. ALL WELDING SHOULD BE SOUND AND COMPLETE WITHOUT ANY GAP, TO AVOID LEAKAGE OF AIR FROM ONE SIDE TO ANOTHER.
02. REFRACTORY TO BE APPLIED BEFORE ERECTING COLD AIR DUCT
03. ANCHOR STUD TO BE WELDED (REF.-038-0002) WITH TUBE SHEET AT GRID.
04. SLEEVES TO BE PROVIDED AT AIR HEATER BLOCK-L AIR ENTRY INSIDE ONLY.
05. SLEEVES ARE TO BE INSERTED & WELDED WITH TUBES AFTER REFRACTORY APPLICATION.
06. ON COMPLETION OF REFRACTORY APPLICATION THE COTTON WASTE SHALL BE REMOVED.
07. AFTER THE INSERTION OF AIR HEATER TUBE, THE SLEEVE WITH ATTACHMENT CLIP TO BE WELDED AT SITE.
08. AIR HEATER TUBES SHALL BE PROTECTS USING END CAPS SUPPLIED ALONG WITH THEM BEFORE APPLICATION OF REFRACTORY.

LIST OF REFERENCE DRAWINGS

DRAWING DOCUMENT ID	DESCRIPTION
JB1014-50050100-038-0002	DETAIL OF AIR HEATER TUBE SHEET
BLS-EPC-001-32-S1	EROSION PROTECTION ATTACHMENT CLIP FOR TUBE $\phi 63.5$ (FOR AIR HEATER TUBES)
BLS-EPS-001-31-S1	EROSION PROTECTION SLEEVE FOR TUBE $\phi 63.5$ (FOR AIR HEATER TUBES)

FOR AIR HEATER TUBES REFER SPEC. DOCUMENT ID - JB1014-50050100-SPC-0001

THIS DRAWING SHOULD BE READ IN CONJUNCTION WITH DWG. No. JB1014-50110107-040-0001 (A & D OF AIR HEATER CASING)

	CASTABLE REFRACTORY FIRE CRETE SUPER	FIRECRETE (SPL)-2500kg/m ³			
---	CASTABLE REFRACTORY	P	B225	--	BOUGHT OUT WILL BE RELEASED IN ELEMENT ID 50140100

ITEM CODE	ITEM DESCRIPTION	LTYP	LGROU	LQTY	LWT (Kg)	REMARK
RPSA202511A	ATTACHMENT CLIP ($\phi 63.5 \times 251g$) P.NO. 5	S	B008	900	63.00	20 Nos. EXTRA (REF.DWG: (BLS-EPC-001-32-S1))
RPSA250011A	HALF SLEEVE ($\phi 63.5 \times 5001g$) P.NO. 4	S	B008	450	585.00	10 Nos. EXTRA (REF.DWG: (BLS-EPS-001-31-S1))

MATERIAL LIST (FOR STANDARD ITEMS)

ITEM CODE	ITEM DESCRIPTION	LTYP	LGROU	LQTY	LWT (Kg)	REMARK
3	TUBE	IS 623 PWF-4, 1982 GR 1	O/D 57.15 X 2.03 THK x 2520 Lg.	467	3245.65	5 QTY EXTRA (SLEEVES FOR AHT001 TUBES)
	AIR HEATER TUBE SLEEVE	P	B008			
2	AIRHEATER TUBE BLOCK-I & II	IS 623 PWF-4, 1982 DW	O/D 63.5 X 2.03 THK x 4496 Lg.	6999	96866.16	ACTUAL+ERECTION SPARE (69 Nos.)
	AIR HEATER TUBE	P	B008			
1	AIRHEATER TUBE BLOCK-I	SA 423 Gr.1ERW	O/D 63.5 X 2.03 THK x 4536 Lg.	2325	32457.00	ACTUAL+ERECTION SPARE (15 Nos.)
	AIR HEATER TUBE	P	B008			
P.NO.	P.DESCRPTION	SPECIFICATION	SIZE	P.QTY	P.WT (kg)	REMARK
ITEM CODE	ITEM DESCRIPTION	LTYP	LGROU	LQTY	LWT (Kg) (TOTAL)	REMARK

CONSULTANT	M/s. AVANT-GARDE ENGINEERS & CONSULTANTS (P) LTD						
CLIENT	M/s. DESCON ENGINEERING PZE (A/c. ETHAD GENERATION POWER LIMITED, PAKISTAN)						
DO NOT SCALE IF IN DOUBT PLEASE ASK	THIS DESIGN IS THE PROPERTY OF M/S. ISGEC HEAVY ENGINEERING LTD. AND IS ONLY ALLOWED TO BE USED BY EXPRESSED PERMISSION AND LICENSE FROM M/S. ISGEC HEAVY ENGINEERING LTD.						
FOR APPROVAL	YES	FOR INFORMATION	NO	FOR PRODUCTION	YES	FOR ERECTION	YES
RESPONSIBLE DEPT.	DRAWN BY SATISH	CHECKED BY ROSHAN LAL	APPROVED BY RAGHVENORA	DATE	03.10.2017		
	TOTAL WEIGHT IN kg.	PROJECT ID JB1014	ELEMENT ID 50050100	I.W.T. NO.	6512		
	648.00	YEAR : 2017	SHEET SIZE : A1	SCALE : 1:40			
 BOILER (formerly Isgec John Thompson) Isgec Heavy Engineering Ltd. Noida 201 301 (U.P.) India		SERVICE	2 x 190 TPH, SINGLE DRUM, WATER TUBE, TOP SUPPORTED, TRAVELLING GRATE BOILER				
		TITLE	GENERAL ARRANGEMENT OF AIR HEATER				
CLIENT:  ETHAD POWER GENERATION LIMITED		DRAWING DOCUMENT ID	JB1014-50050100-038-0001 REVISION 00				
EPC CONSULTANT							
 AVANT-GARDE ENGINEERS & CONSULTANTS PVT. LIMITED							
EPC CONTRACTOR							
 DESCON ENGINEERING LIMITED DESCON HEADQUARTERS 18-KM FERROZPUR ROAD,P.O.BOX 1201,LAHORE-53000 PAKISTAN TEL:+92(42)5990034, FAX :+92(42)5811005-5811135 WEBSITE: www.descon.com.pk, EMAIL:descon@descon.com.pk							
PROJECT NAME & LOCATION: 2x37.2 MW Cogenproject Etihad Power Limited Rahim Yar Khan, Punjab Pakistan							
DRAWING TITLE: GENERAL ARRANGEMENT OF AIR HEATER							
SHEET SIZE	SCALE	DRAWING #.	PROJECT NO.	SHEETS	REV.		
A1	1:40	V-4574-681-D-268-00	22001	NO. TOTAL			
			1	1			